

GABLS1: 128³ resolution study

GABLS1 32-cubed to 128-cubed resolution sensitivity | 24 September 2026. Matched nine-hour WENO9 cases use a 400 m cube, seed 123, paired initial perturbations, filtered rough-wall drag, and gamma=2 scheme-native SLD with a 300 s filter. The fixed 1 m LES medians are intercomparison references, not observations. The 128-cubed grid spacing is 3.125 m. Shear below is computed on common 12.5 m intervals after interpolating both u and v to 32-cubed cell centers. Native shear is shown separately.

During 8-9 h, one-face SLD shear at 12.5, 25, and 37.5 m is 0.09152/0.06140/0.05235 per second at 32 cubed, 0.10144/0.05600/0.03951 at 64 cubed, and 0.09080/0.04759/0.03967 at 128 cubed. The fixed 1 m median is 0.10038/0.06666/0.05607. The 128-cubed two-face result is 0.09242/0.04739/0.04083. The third grid does not establish convergence or eliminate the upper-shear deficit. Both 7-8 h and 8-9 h windows appear in the figures.

Peak resolved w-squared in 8-9 h is 0.05588/0.09688/0.11556 square meters per second squared for 32/64/128-cubed one-face SLD; 128-cubed no closure gives 0.11063, and two-face SLD gives 0.11171. Resolved turbulence remains grid-sensitive. At 18.75 m, interpolated w-cubed is +0.002489 and +0.002144 cubic meters per second cubed for 128 one/two-face, compared with negative values at 64 cubed. No w-cubed median is available, so the sign change is sensitivity, not a fidelity ranking.

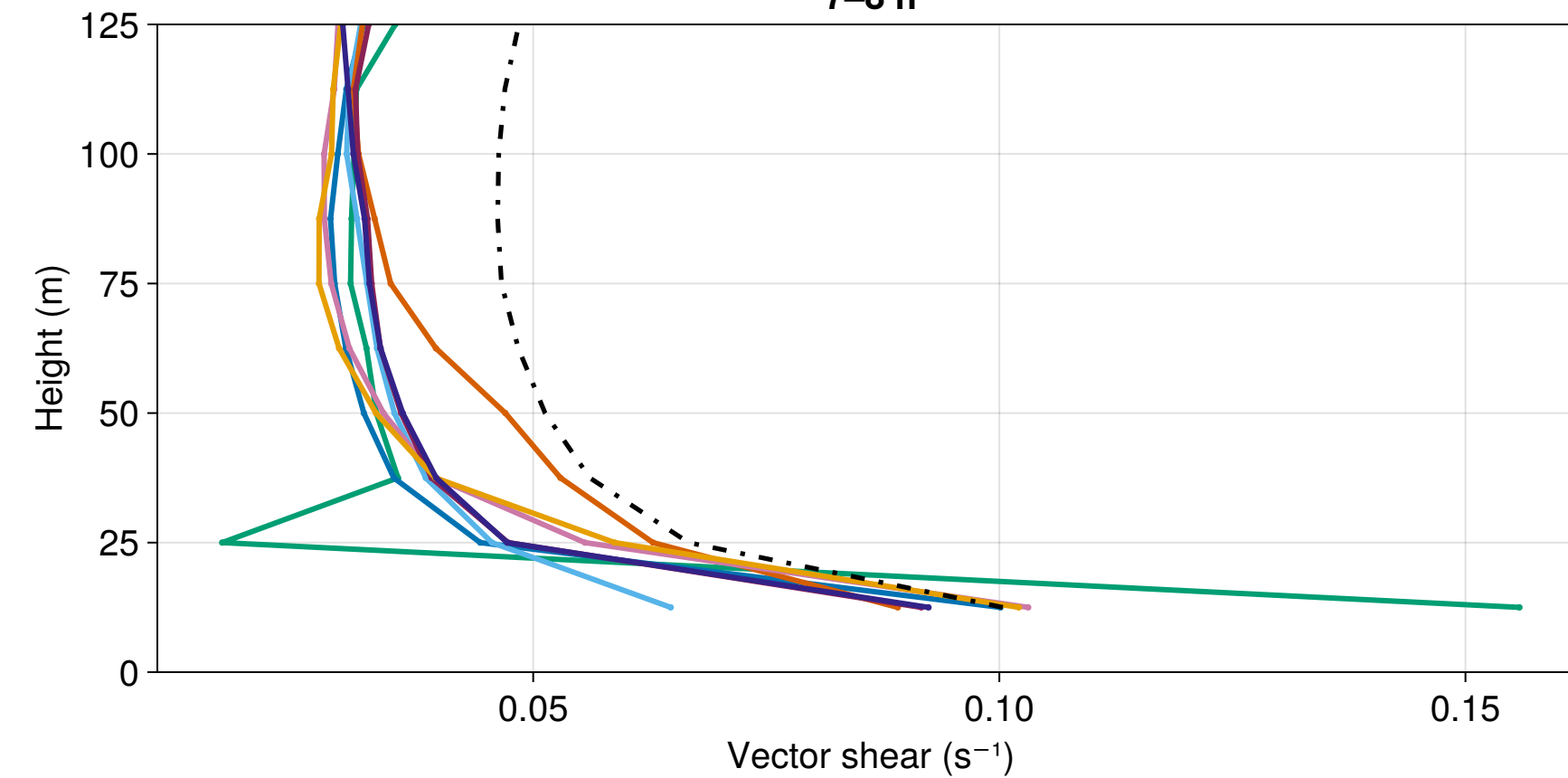
The one-face closure reaches $z=12.5/6.25/3.125$ m at 32/64/128 cubed. Two-face support reaches 12.5 m at 64 cubed but only 6.25 m at 128 cubed. The current implementation supports at most two faces; none of the 128-cubed cases retains the 12.5 m physical footprint. Refinement therefore changes closure reach as well as grid spacing. At 128 face two, mean viscosity is 0.04693 square meters per second, resolved u-w is -0.05477, SGS u-w is -0.00619, and the separate WENO correction is -0.01079 square meters per second squared.

Validation: immutable 128-cubed source-manifest SHA-256
dfdcd8846a756b73b7006b67a0bdcd565a2df5400bf769e64f9d7d5345e3e233. H100 gate 7674 passed CUDA parity34/34 and 15/15, 32/32, 38/38 short-case checks. Science array7688 tasks1-3 each ended with durable exit zero and CASE_DONE at 32400 s. Independent Julia audits verified source, paired initial state, native128/129 levels, exact schedules, finite fields, evolving wall filter, density-consistent surface flux, and scheme-native flux identities. Both SLD cases passed 55 local1/L and stability records through nine hours. An audit-only face-two height was corrected to 6.25 m; the frozen simulation was unchanged. Compact exports, scripts and evidence accompany this chapter. Three grids show a trend, not a formal convergence proof.

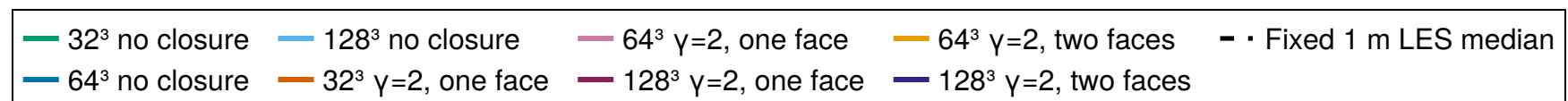
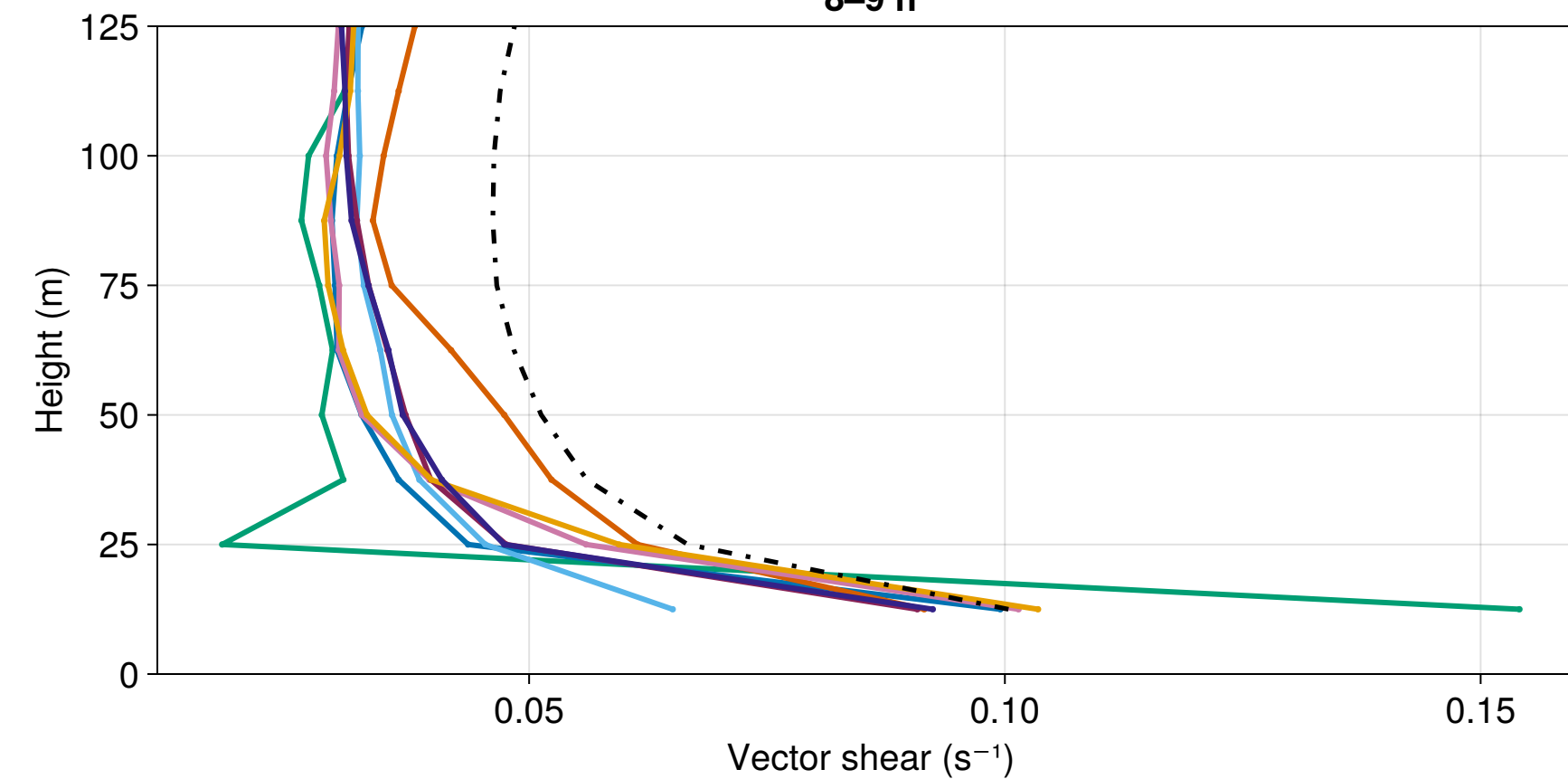
Vector shear at matched physical heights

u and v interpolated to 32^3 cell centers (6.25, 18.75, ... m), then differenced

7–8 h



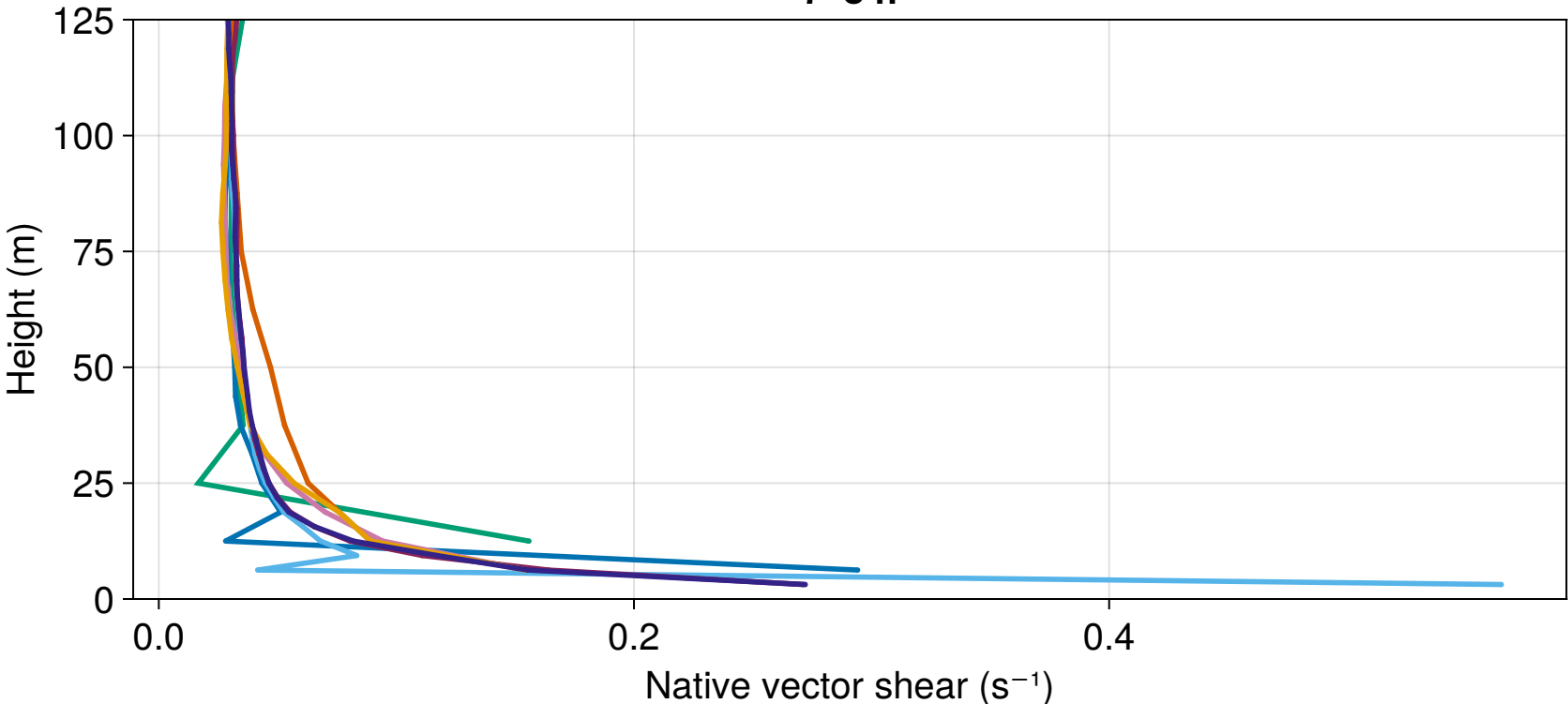
8–9 h



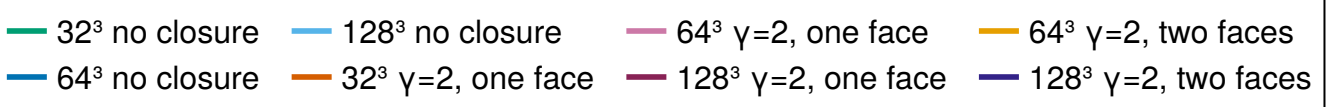
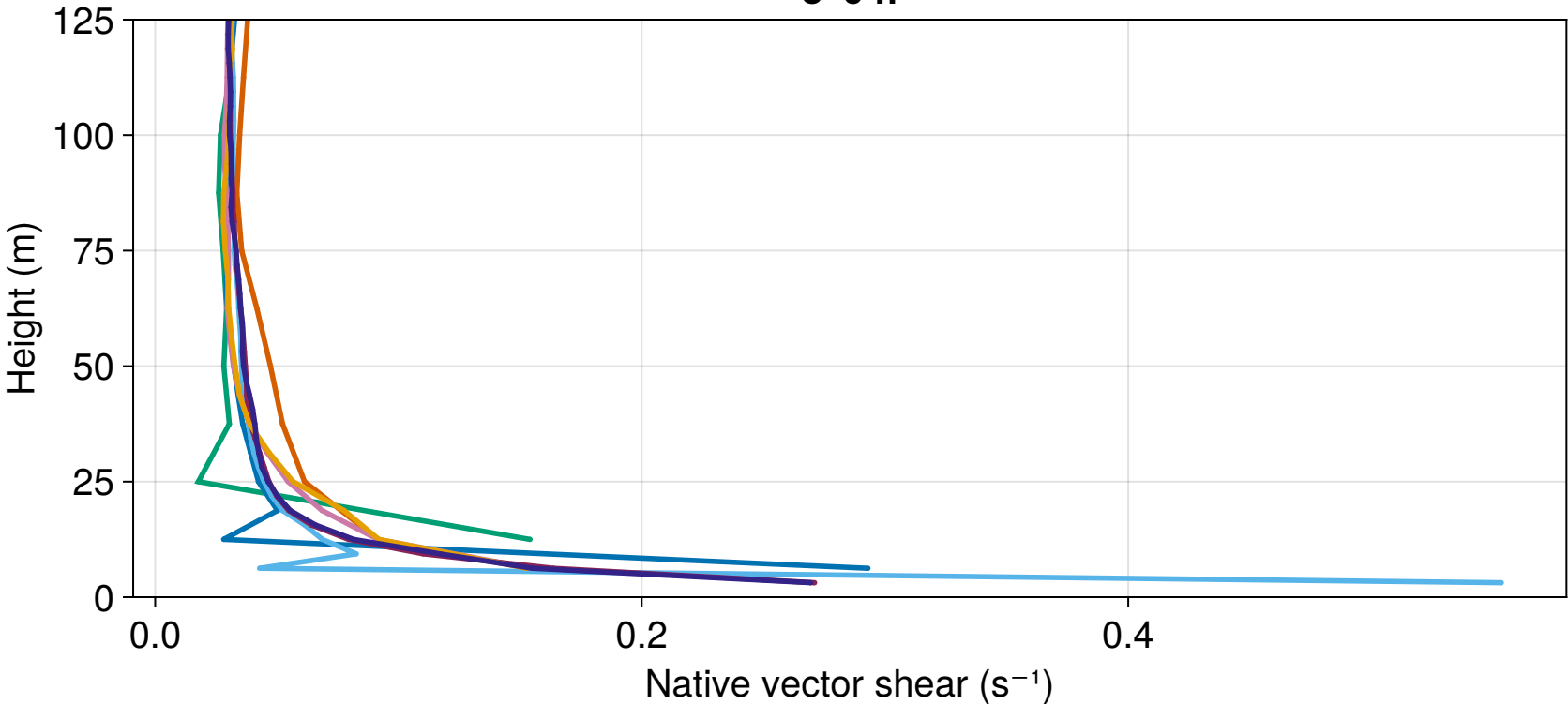
Native-grid vector shear

128³/64³/32³ native differences span 3.125/6.25/12.5 m; compare common-height shear.

7–8 h

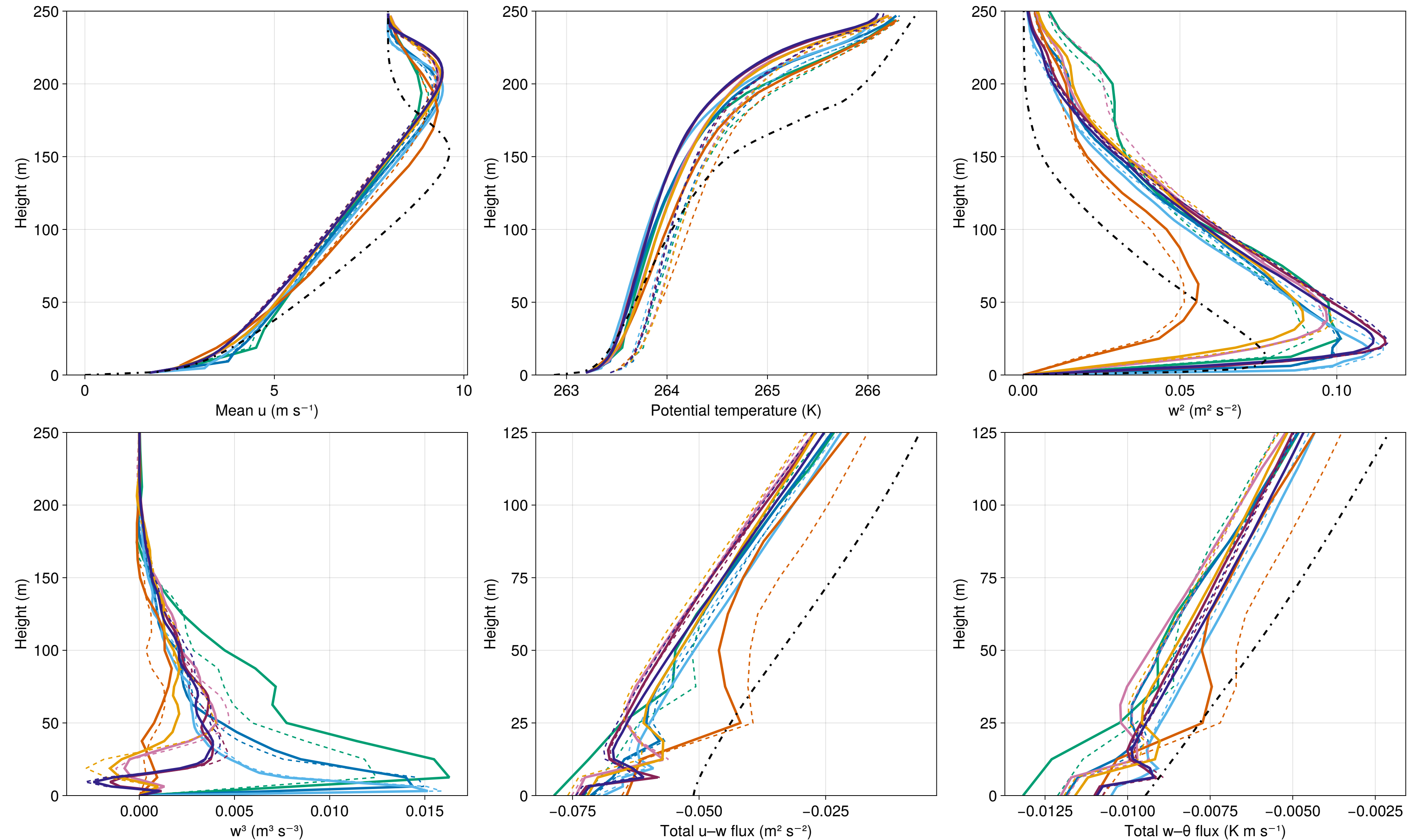


8–9 h



GABLS1 $32^3 \rightarrow 64^3 \rightarrow 128^3$: matched stability-corrected SLD

400 m cube · WENO9 · $\gamma=2$ · 300 s wall/SLD filters · dashed 7–8 h, solid 8–9 h

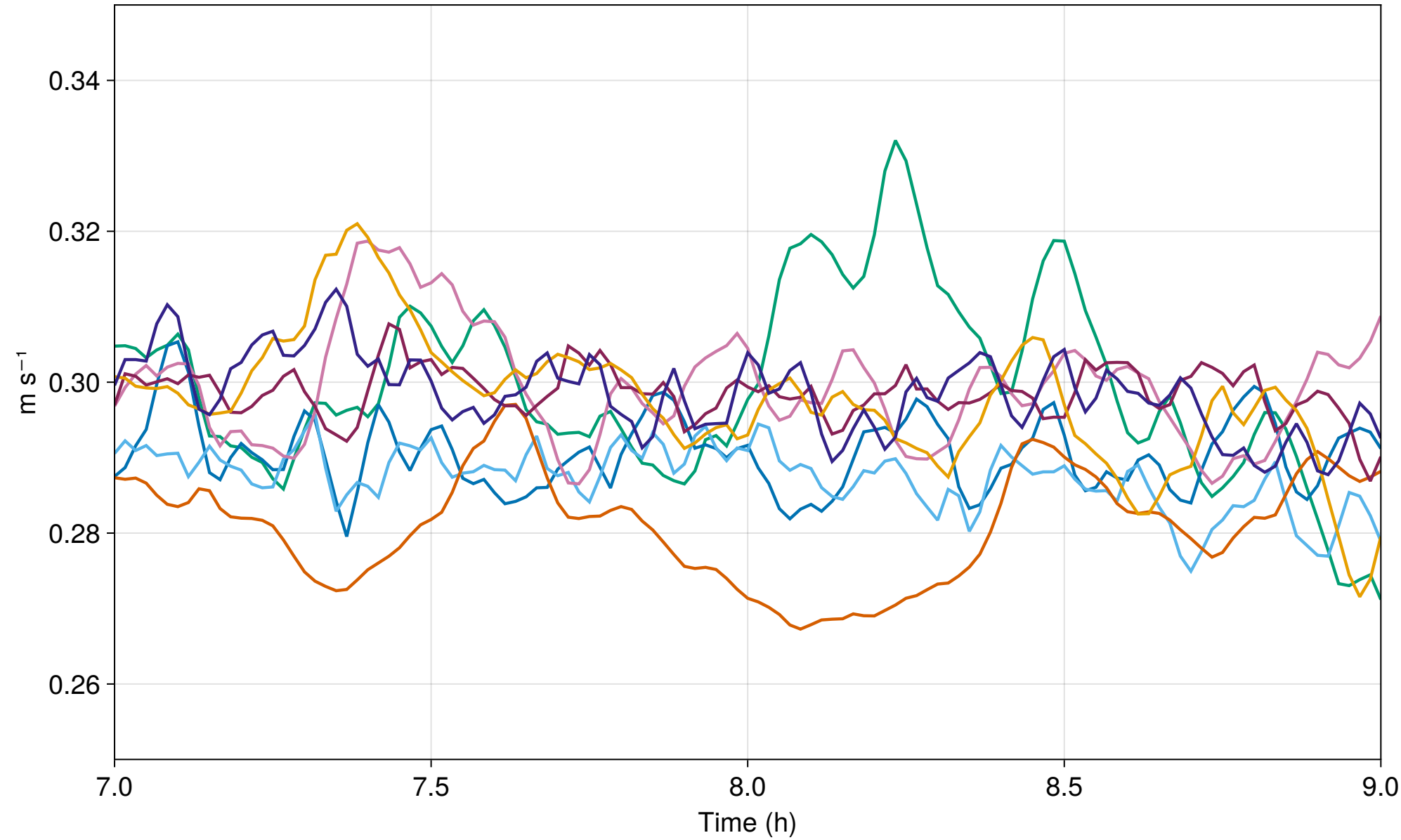


— 32^3 no closure
 — 128^3 no closure
 — 64^3 $\gamma=2$, one face
 — 64^3 $\gamma=2$, two faces
 - - Fixed 1 m LES median
— 64^3 no closure
 — 32^3 $\gamma=2$, one face
 — 128^3 $\gamma=2$, one face
 — 128^3 $\gamma=2$, two faces

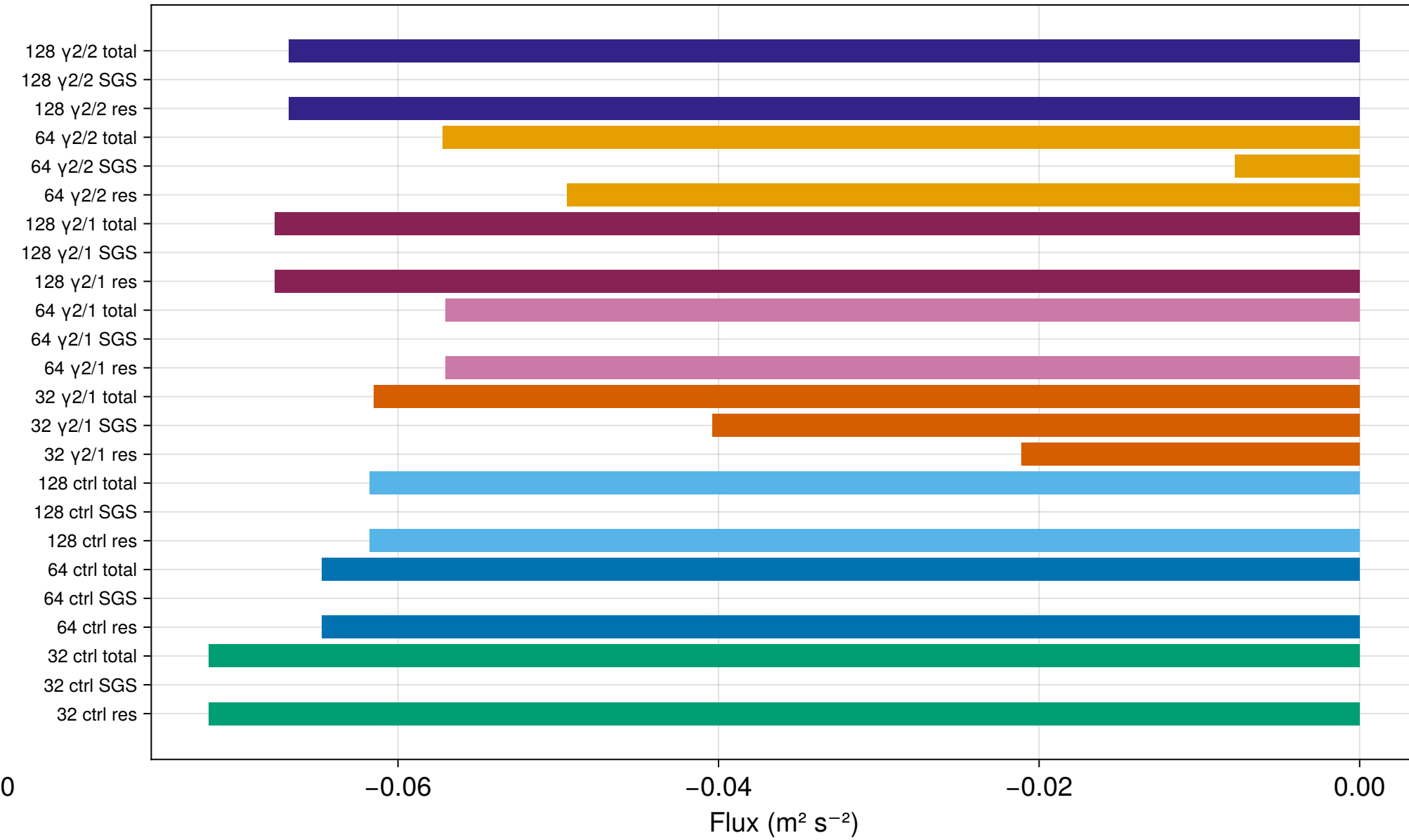
Wall response and vertical-transport partition

Common face $z=12.5$ m · 128³ SLD support ends at 3.125 or 6.25 m

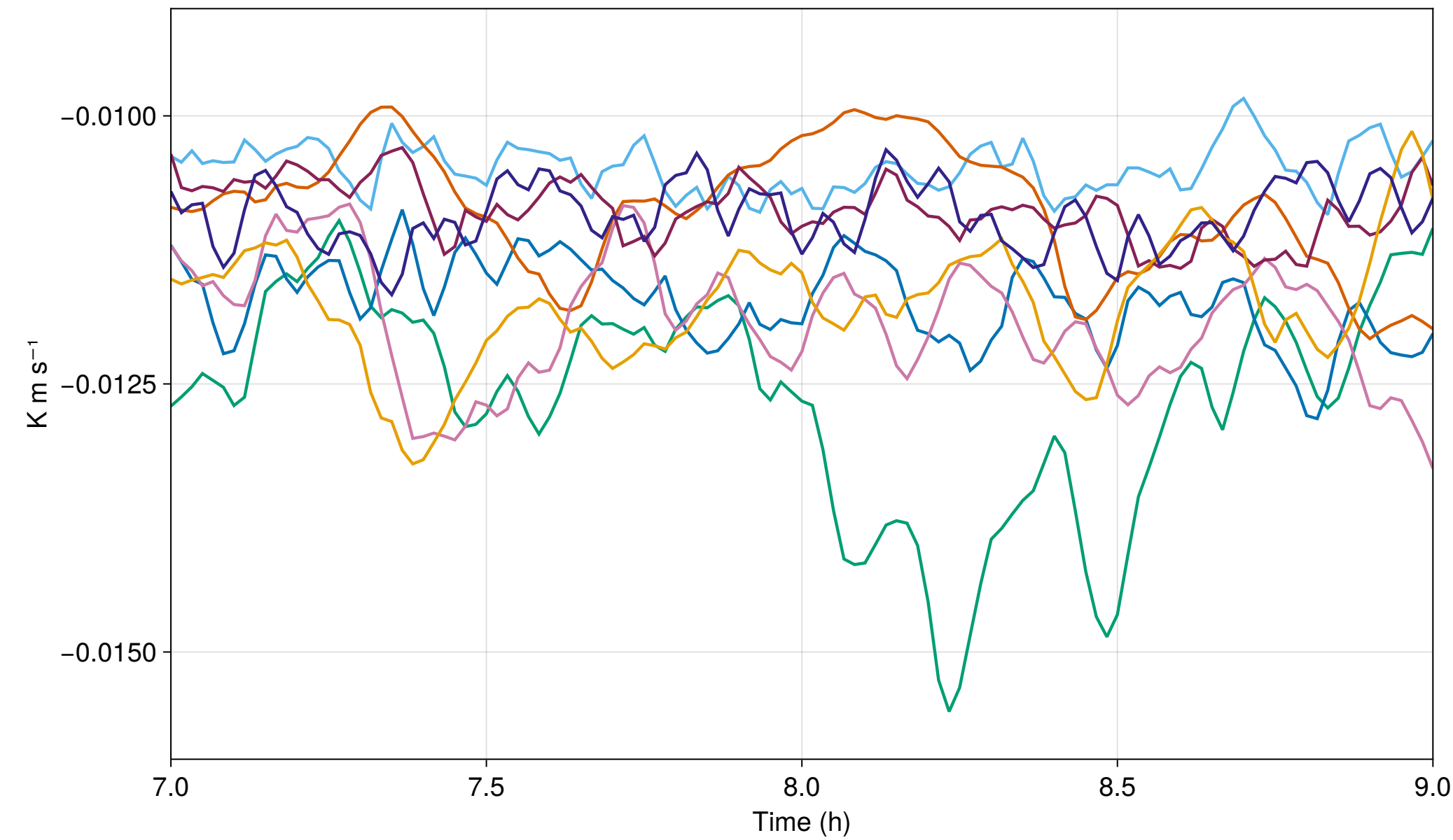
Friction velocity



$u-w$ at $z=12.5$ m



Surface heat flux



$w-\theta$ at $z=12.5$ m

